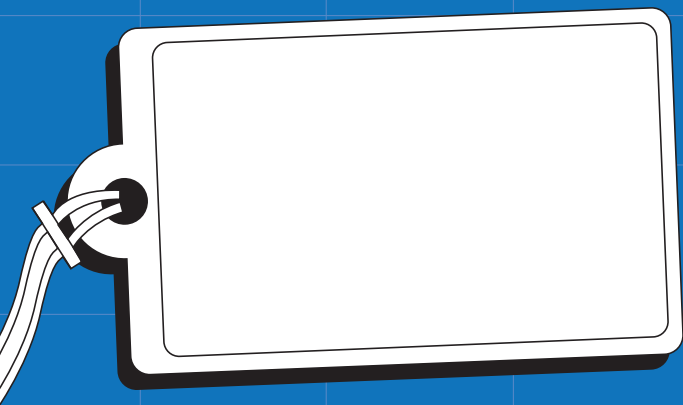




SURDS
&
INDICES

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Solutions

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MATHLETICS

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Page 3 questions

Why are ALL integers rational numbers? Give an example.

A rational number is one that can be written as a fraction with integers in the numerator and denominator. Every integer like -1023 can be written as a fraction with 1 in the denominator, like $\frac{-1023}{1}$. This means that all integers are rational.

Identify whether these numbers are rational or irrational. Give a reason.

- a $\frac{5}{11}$ is a rational number as it is a fraction with integers in the numerator and denominator.
- b 10 is rational as it can be expressed as $\frac{10}{1}$
- c 5.3472294401... is an irrational number as is a non repeating decimal with no end.
- d 0.3333333... is a rational number. Although it does not have an end it does repeat. It can be written as $\frac{3}{9} = \frac{1}{3}$.
- e 0.324324324324324 is a repeating infinite decimal, so it is rational. It can be written as $\frac{324}{999}$.
Try it with a calculator.
- f 0.43554 is a finite decimal (it has an end), so it is rational. It can be written as $\frac{43554}{100000}$

Identify whether each of the following numbers is rational, irrational or is undefined (no real solution). Give a reason.

a $\sqrt{-10}$

$\sqrt{-10}$ is undefined as there is no real number that can be multiplied by itself to give -10 .

b $\sqrt{12}$

$$\sqrt{12} = \sqrt{4 \times 3} = 2\sqrt{3} = 3.46410161514...$$

$\sqrt{3}$ is an infinite non recurring decimal, so $\sqrt{12}$ is an irrational number.

c $\sqrt{9}$

$$\sqrt{9} = \sqrt{3^2} = 3$$

3 is an integer so $\sqrt{9}$ is a rational number.

Page 3 questions

d $\sqrt{100}$

$$\sqrt{100} = 10$$

10 is an integer so $\sqrt{100}$ is a rational number.

e $\sqrt{5}$

$\sqrt{5} = 2.2360679775...$ which is an infinite non recurring decimal, so $\sqrt{5}$ is an irrational number.

f $\sqrt{36}$

$\sqrt{36} = 6$ which is an integer, so $\sqrt{36}$ is a rational number.

g $\sqrt{0}$

$\sqrt{0} = 0 = \frac{0}{102221}$ which makes it a rational number.

h $\sqrt{-100}$

$\sqrt{-100}$ is undefined as there is no real number that can be multiplied by itself to give -100 .

Between which two integers are these surds?

a $\sqrt{11}$

$$\sqrt{9} < \sqrt{11} < \sqrt{16}$$

$$3 < \sqrt{11} < 4$$

so $\sqrt{11}$ is between 3 and 4.

b $\sqrt{40}$

$$\sqrt{36} < \sqrt{40} < \sqrt{49}$$

$$6 < \sqrt{40} < 7$$

so $\sqrt{40}$ is between 6 and 7.

c $\sqrt{111}$

$$\sqrt{100} < \sqrt{111} < \sqrt{121}$$

$$10 < \sqrt{111} < 11$$

so $\sqrt{111}$ is between 10 and 11.

d $\sqrt{93}$

$$\sqrt{81} < \sqrt{93} < \sqrt{100}$$

$$9 < \sqrt{93} < 10$$

so $\sqrt{93}$ is between 9 and 10.

Page 7 questions

Find the following products. Simplify to an integer if possible:

a $\sqrt{2} \times \sqrt{8}$

$$= \sqrt{16}$$

$$= 4$$

b $(\sqrt{11})^2$

$$= \sqrt{11} \times \sqrt{11}$$

$$= 11$$

c $\sqrt{18} \times \sqrt{8}$

$$= \sqrt{144}$$

$$= 12$$

d $\sqrt{5} \times \sqrt{6}$

$$= \sqrt{30}$$

e $\sqrt{5} \times \sqrt{10} \times \sqrt{2}$

$$= \sqrt{100}$$

$$= 10$$

f $\sqrt{3} \times \sqrt{6} \times \sqrt{5}$

$$= \sqrt{90}$$

$$= \sqrt{9} \times \sqrt{10}$$

$$= 3\sqrt{10}$$

Find the following quotients. Simplify to an integer if possible:

a $\sqrt{\frac{144}{9}}$

$$= \frac{\sqrt{144}}{\sqrt{9}}$$

$$= \frac{12}{3}$$

$$= 4$$

b $\sqrt{\frac{121}{9}}$

$$= \frac{\sqrt{121}}{\sqrt{9}}$$

$$= \frac{11}{3}$$

c $\sqrt{\frac{100}{20}}$

$$= \sqrt{5}$$

d $\sqrt{\frac{3}{64}}$

$$= \frac{\sqrt{3}}{\sqrt{64}}$$

$$= \frac{\sqrt{3}}{8}$$

Page 7 questions

e $\sqrt{\frac{100}{49}}$

$$= \frac{\sqrt{100}}{\sqrt{49}}$$

$$= \frac{10}{7}$$

f $\sqrt{\frac{8}{81}}$

$$= \frac{\sqrt{8}}{\sqrt{81}}$$

$$= \frac{\sqrt{4} \times \sqrt{2}}{9}$$

$$= \frac{2\sqrt{2}}{9}$$

Page 8 questions

Write the following in simplest surd form:

a $\sqrt{18}$

$$= \sqrt{9 \times 2}$$

$$= \sqrt{9} \times \sqrt{2}$$

$$= 3 \times \sqrt{2}$$

$$= 3\sqrt{2}$$

b $\sqrt{24}$

$$= \sqrt{4 \times 6}$$

$$= \sqrt{4} \times \sqrt{6}$$

$$= 2\sqrt{6}$$

c $\sqrt{80}$

$$= \sqrt{16 \times 5}$$

$$= \sqrt{16} \times \sqrt{5}$$

$$= 4\sqrt{5}$$

d $\sqrt{150}$

$$= \sqrt{25 \times 6}$$

$$= \sqrt{25} \times \sqrt{6}$$

$$= 5\sqrt{6}$$

e $\sqrt{1200}$

$$= \sqrt{400 \times 3}$$

$$= \sqrt{400} \times \sqrt{3}$$

$$= 20\sqrt{3}$$

f $\sqrt{147}$

$$= \sqrt{49 \times 3}$$

$$= \sqrt{49} \times \sqrt{3}$$

$$= 7\sqrt{3}$$

Page 8 questions

Simplify the following into simplest surd form:

a $\sqrt{10} \times \sqrt{8}$

Two approaches are possible.
The result is the same.

$$\begin{aligned} \text{either } & \sqrt{10} \times \sqrt{8} \\ &= \sqrt{10 \times 8} \\ &= \sqrt{80} \\ &= \sqrt{16 \times 5} \\ &= 4\sqrt{5} \end{aligned}$$

$$\begin{aligned} \text{or } & \sqrt{10} \times \sqrt{8} \\ &= \sqrt{5 \times 2} \times \sqrt{4 \times 2} \\ &= \sqrt{5} \times \sqrt{2} \times \sqrt{4} \times \sqrt{2} \\ &= \sqrt{5} \times \sqrt{2} \times 2 \times \sqrt{2} \\ &= 2 \times \sqrt{5} \times \sqrt{2} \times \sqrt{2} \\ &= 2\sqrt{5} \times 2 \\ &= 4\sqrt{5} \end{aligned}$$

d $4\sqrt{10} \times 2\sqrt{5}$

$$\begin{aligned} &= (4 \times 2) \times (\sqrt{10} \times \sqrt{5}) \\ &= 8\sqrt{50} \\ &= 8\sqrt{25 \times 2} \\ &= 8 \times \sqrt{25} \times \sqrt{2} \\ &= 8 \times 5 \times \sqrt{2} \\ &= 40\sqrt{2} \end{aligned}$$

e $3\sqrt{\frac{9}{4}}$

$$\begin{aligned} &= 3 \times \frac{\sqrt{9}}{\sqrt{4}} \\ &= 3 \times \frac{3}{2} \\ &= \frac{9}{2} \end{aligned}$$

b $\sqrt{5} \times \sqrt{40}$

$$\begin{aligned} &= \sqrt{200} \\ &= \sqrt{100 \times 2} \\ &= \sqrt{100} \times \sqrt{2} \\ &= 10\sqrt{2} \end{aligned}$$

c $2\sqrt{2} \times \sqrt{75}$

$$\begin{aligned} &= 2\sqrt{2} \times \sqrt{25 \times 3} \\ &= 2\sqrt{2} \times 5\sqrt{3} \\ &= 10\sqrt{6} \end{aligned}$$

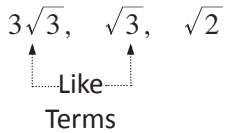
f $6\sqrt{25} \div 3\sqrt{12}$

$$\begin{aligned} &= \frac{6\sqrt{25}}{3\sqrt{12}} \\ &= \frac{6 \times \sqrt{25}}{3 \times \sqrt{12}} \\ &= \frac{2 \times 5}{\sqrt{12}} \\ &= \frac{2 \times 5}{\sqrt{4 \times 3}} \\ &= \frac{2 \times 5}{2 \times \sqrt{3}} \\ &= \frac{5}{\sqrt{3}} \\ &= \frac{5}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} \\ &= \frac{5\sqrt{3}}{3} \end{aligned}$$

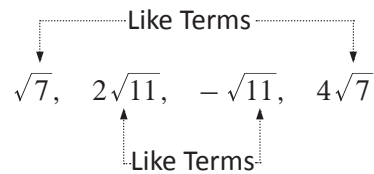
Page 9 questions

Find the like terms in each of the following:

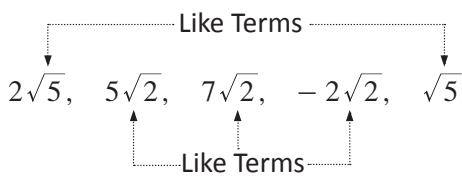
a $3\sqrt{3}, \sqrt{3}, \sqrt{2}$



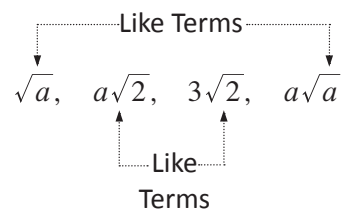
b $\sqrt{7}, 2\sqrt{11}, -\sqrt{11}, 4\sqrt{7}$



c $2\sqrt{5}, 5\sqrt{2}, 7\sqrt{2}, -2\sqrt{2}, \sqrt{5}$



d $\sqrt{a}, a\sqrt{2}, 3\sqrt{2}, a\sqrt{a}$



Simplify these expressions, if possible:

a $7\sqrt{3} - 5\sqrt{3}$

$= 2\sqrt{3}$

b $2\sqrt{5} + 4\sqrt{5} - 7\sqrt{5}$

$= 6\sqrt{5} - 7\sqrt{5}$

$= -1\sqrt{5}$

$= -\sqrt{5}$

c $6\sqrt{2} + 5\sqrt{3} - 4\sqrt{2} - 7\sqrt{3}$

$= 2\sqrt{2} - 2\sqrt{3}$

d $3\sqrt{5} + 5\sqrt{3} - 2\sqrt{5} + 4\sqrt{5}$

$= \sqrt{5}(3 - 2 + 4) + 5\sqrt{3}$

$= 5\sqrt{5} + 5\sqrt{3}$

Page 9 questions

Find the simplest surd form and then simplify the following as much as possible:

a $\sqrt{18} - \sqrt{2} + \sqrt{50}$

$$\begin{aligned} &= \sqrt{9 \times 2} - \sqrt{2} + \sqrt{25 \times 2} \\ &= 3\sqrt{2} - \sqrt{2} + 5\sqrt{2} \\ &= 7\sqrt{2} \end{aligned}$$

b $\sqrt{108} + \sqrt{75} - \sqrt{48}$

$$\begin{aligned} &= \sqrt{36 \times 3} + \sqrt{25 \times 3} - \sqrt{16 \times 3} \\ &= 6\sqrt{3} + 5\sqrt{3} - 4\sqrt{3} \\ &= 7\sqrt{3} \end{aligned}$$

c $\sqrt{112} - \sqrt{6} - \sqrt{7} + \sqrt{150}$

$$\begin{aligned} &= \sqrt{16 \times 7} - \sqrt{6} - \sqrt{7} + \sqrt{25 \times 6} \\ &= 4\sqrt{7} - \sqrt{6} - \sqrt{7} + 5\sqrt{6} \\ &= 3\sqrt{7} + 4\sqrt{6} \end{aligned}$$

d $\sqrt{40} + \sqrt{1000} - \sqrt{20} - \sqrt{10}$

$$\begin{aligned} &= \sqrt{4 \times 10} + \sqrt{100 \times 10} - \sqrt{2 \times 10} - \sqrt{10} \\ &= 2\sqrt{10} + 10\sqrt{10} - \sqrt{2}\sqrt{10} - \sqrt{10} \\ &= \sqrt{10}(2 + 10 - \sqrt{2} - 1) \\ &= \sqrt{10}(11 - \sqrt{2}) \end{aligned}$$

Page 12 questions

Simplify these products to simplest surd form:

a $\sqrt{5}(2\sqrt{3} + \sqrt{7})$

$$\begin{aligned} &= \sqrt{5} \times 2\sqrt{3} + \sqrt{5}\sqrt{7} \\ &= 2\sqrt{15} + \sqrt{35} \end{aligned}$$

b $\sqrt{3}(2\sqrt{2} - 7\sqrt{5})$

$$\begin{aligned} &= \sqrt{3} \times 2\sqrt{2} - \sqrt{3} \times 7\sqrt{5} \\ &= 2\sqrt{6} - 7\sqrt{15} \end{aligned}$$

c $\sqrt{11}(3\sqrt{11} + 4\sqrt{2})$

$$\begin{aligned} &= \sqrt{11} \times 3\sqrt{11} + \sqrt{11} \times 4\sqrt{2} \\ &= 3 \times 11 + 4\sqrt{22} \\ &= 33 + 4\sqrt{22} \end{aligned}$$

d $3\sqrt{10}(6\sqrt{2} - 8\sqrt{7})$

$$\begin{aligned} &= 3\sqrt{10} \times 6\sqrt{2} - 3\sqrt{10} \times 8\sqrt{7} \\ &= 18\sqrt{20} - 24\sqrt{70} \end{aligned}$$

e $3\sqrt{7}(7\sqrt{3} + 2\sqrt{15})$

$$\begin{aligned} &= 3\sqrt{7} \times 7\sqrt{3} + 3\sqrt{7} \times 2\sqrt{15} \\ &= 21\sqrt{21} + 6\sqrt{105} \end{aligned}$$

f $2\sqrt{2}(3\sqrt{32} + \sqrt{72} - 2\sqrt{7})$

$$\begin{aligned} &= 2\sqrt{2} \times 3\sqrt{32} + 2\sqrt{2} \times \sqrt{72} - 2\sqrt{2} \times 2\sqrt{7} \\ &= 6\sqrt{64} + 2\sqrt{144} - 4\sqrt{14} \\ &= 6 \times 8 + 2 \times 12 - 4\sqrt{14} \\ &= 48 + 24 - 4\sqrt{14} \\ &= 72 - 4\sqrt{14} \end{aligned}$$

g $\sqrt{x}(3\sqrt{2} + 2\sqrt{5})$

$$\begin{aligned} &= \sqrt{x} \times 3\sqrt{2} + \sqrt{x} \times 2\sqrt{5} \\ &= 3\sqrt{2x} + 2\sqrt{5x} \end{aligned}$$

h $\sqrt{p}(2\sqrt{p} - 4\sqrt{10})$

$$\begin{aligned} &= \sqrt{p} \times 2\sqrt{p} - \sqrt{p} \times 4\sqrt{10} \\ &= 2p - 4\sqrt{10p} \end{aligned}$$

i $2\sqrt{a}(5\sqrt{6} - \sqrt{a})$

$$\begin{aligned} &= 2\sqrt{a} \times 5\sqrt{6} - 2\sqrt{a} \times \sqrt{a} \\ &= 10\sqrt{6a} - 2a \end{aligned}$$

j $3\sqrt{m}(m\sqrt{3} + 2\sqrt{m})$

$$\begin{aligned} &= 3\sqrt{m} \times m\sqrt{3} + 3\sqrt{m} \times 2\sqrt{m} \\ &= 3m\sqrt{3m} + 6m \end{aligned}$$

Page 13 questions

Find these binomial products and simplify to simplest surd form:

a $(\sqrt{2} + \sqrt{3})(\sqrt{7} + \sqrt{6})$

$$\begin{aligned} &= \sqrt{2} \times \sqrt{7} + \sqrt{2} \times \sqrt{6} + \sqrt{3} \times \sqrt{7} + \sqrt{3} \times \sqrt{6} \\ &= \sqrt{14} + \sqrt{12} + \sqrt{21} + \sqrt{18} \end{aligned}$$

b $(4\sqrt{3} - \sqrt{5})(\sqrt{32} + \sqrt{20})$

$$\begin{aligned} &= (4\sqrt{3} - \sqrt{5})(4\sqrt{2} + 2\sqrt{5}) \\ &= 4\sqrt{3} \times 4\sqrt{2} + 4\sqrt{3} \times 2\sqrt{5} - \sqrt{5} \times 4\sqrt{2} - \sqrt{5} \times 2\sqrt{5} \\ &= 16\sqrt{6} + 8\sqrt{15} + 4\sqrt{10} - 10 \end{aligned}$$

c $(3\sqrt{7} - 2\sqrt{10})(\sqrt{5} + \sqrt{11})$

$$\begin{aligned} &= 3\sqrt{7} \times \sqrt{5} + 3\sqrt{7} \times \sqrt{11} - 2\sqrt{10} \times \sqrt{5} - 2\sqrt{10} \times \sqrt{11} \\ &= 3\sqrt{35} + 3\sqrt{77} - 2\sqrt{50} - 2\sqrt{55} \\ &= 3\sqrt{35} + 3\sqrt{77} - 10\sqrt{2} - 2\sqrt{55} \end{aligned}$$

d $(\sqrt{7} + 5\sqrt{2})(6\sqrt{5} - \sqrt{3})$

$$\begin{aligned} &= \sqrt{7} \times 6\sqrt{5} - \sqrt{7} \times \sqrt{3} + 5\sqrt{2} \times 6\sqrt{5} - 5\sqrt{2} \times \sqrt{3} \\ &= 6\sqrt{35} - \sqrt{21} + 30\sqrt{10} - 5\sqrt{6} \end{aligned}$$

e $(8\sqrt{5} - 3\sqrt{2})(7\sqrt{10} - 4\sqrt{2})$

$$\begin{aligned} &= 8\sqrt{5} \times 7\sqrt{10} - 8\sqrt{5} \times 4\sqrt{2} - 3\sqrt{2} \times 7\sqrt{10} + 3\sqrt{2} \times 4\sqrt{2} \\ &= 56\sqrt{50} - 32\sqrt{10} - 21\sqrt{20} + 12\sqrt{4} \\ &= 280\sqrt{2} - 24\sqrt{10} - 21\sqrt{20} + 24 \end{aligned}$$

Page 13 questions

f $(3\sqrt{6}7\sqrt{5})(2\sqrt{11} - 6\sqrt{5})$

$$\begin{aligned} &= 3\sqrt{6} \times 2\sqrt{11} - 3\sqrt{6} \times 6\sqrt{5} + 7\sqrt{5} \times 2\sqrt{11} - 7\sqrt{5} \times 6\sqrt{5} \\ &= 6\sqrt{66} - 18\sqrt{30} + 14\sqrt{55} - 42\sqrt{25} \\ &= 6\sqrt{66} - 18\sqrt{30} + 14\sqrt{55} - 210 \end{aligned}$$

g $(\sqrt{5} + \sqrt{7})^2$

$$\begin{aligned} &= (\sqrt{5} + \sqrt{7})(\sqrt{5} + \sqrt{7}) \\ &= \sqrt{5} \times \sqrt{5} + \sqrt{5} \times \sqrt{7} + \sqrt{7} \times \sqrt{5} + \sqrt{7} \times \sqrt{7} \\ &= 5 + \sqrt{35} + \sqrt{35} + 7 \\ &= 12 + 2\sqrt{35} \end{aligned}$$

h $(2\sqrt{3} - 11\sqrt{5})^2$

$$\begin{aligned} &= (2\sqrt{3} - 11\sqrt{5})(2\sqrt{3} - 11\sqrt{5}) \\ &= 2\sqrt{3} \times 2\sqrt{3} - 2\sqrt{3} \times 11\sqrt{5} - 11\sqrt{5} \times 2\sqrt{3} + 11\sqrt{5} \times 11\sqrt{5} \\ &= 4 \times 3 - 22\sqrt{15} - 22\sqrt{15} + 121 \times 5 \\ &= 12 - 22\sqrt{15} + 605 \\ &= 617 - 22\sqrt{15} \end{aligned}$$

i $(2\sqrt{3} - 5\sqrt{11})(2\sqrt{3} + 5\sqrt{11})$

$$\begin{aligned} &= 2\sqrt{3} \times 2\sqrt{3} + 2\sqrt{3} \times 5\sqrt{11} - 5\sqrt{11} \times 2\sqrt{3} - 5\sqrt{11} \times 5\sqrt{11} \\ &= 4 \times 3 + 10\sqrt{33} - 10\sqrt{33} - 25 \times 11 \\ &= 12 - 275 \\ &= -263 \end{aligned}$$

j $(4\sqrt{x} + 3\sqrt{3})(4\sqrt{x} - 3\sqrt{3})$

$$\begin{aligned} &= 4\sqrt{x} \times 4\sqrt{x} - 4\sqrt{x} \times 3\sqrt{3} + 3\sqrt{3} \times 4\sqrt{x} - 3\sqrt{3} \times 3\sqrt{3} \\ &= 16x - 12\sqrt{3x} + 12\sqrt{3x} - 9 \times 3 \\ &= 16x - 27 \end{aligned}$$

Page 14 questions

Write down the conjugate of the following:

a $2\sqrt{5} + \sqrt{6}$

$2\sqrt{5} - \sqrt{6}$

b $\sqrt{10} - 3\sqrt{7}$

$\sqrt{10} + 3\sqrt{7}$

c $5\sqrt{11} - 2\sqrt{2}$

$5\sqrt{11} + 2\sqrt{2}$

d $6\sqrt{7} + 8\sqrt{10}$

$6\sqrt{7} - 8\sqrt{10}$

Find the product of the following with its conjugate:

For each example use the fact that the product of a term with its conjugate is the square of the second term subtracted from the square of the first term, that is $(a + b)(a - b) = a^2 - b^2$.

a $4 - \sqrt{5}$

$$\begin{aligned}\text{product} &= a^2 - b^2 \\ &= 4^2 - \sqrt{5}^2 \\ &= 16 - 5 \\ &= 9\end{aligned}$$

b $\sqrt{3} + \sqrt{6}$

$$\begin{aligned}\text{product} &= a^2 - b^2 \\ &= \sqrt{3}^2 - \sqrt{6}^2 \\ &= 3 - 6 \\ &= -3\end{aligned}$$

c $5\sqrt{3} + \sqrt{7}$

$$\begin{aligned}\text{product} &= a^2 - b^2 \\ &= (5\sqrt{3})^2 - \sqrt{7}^2 \\ &= 25 \times 3 - 7 \\ &= 75 - 7 \\ &= 68\end{aligned}$$

d $6\sqrt{2} - 3\sqrt{10}$

$$\begin{aligned}\text{product} &= a^2 - b^2 \\ &= (6\sqrt{2})^2 - (3\sqrt{10})^2 \\ &= 36 \times 2 - 9 \times 10 \\ &= 72 - 90 \\ &= -18\end{aligned}$$

e $2\sqrt{11} + 4\sqrt{15}$

$$\begin{aligned}\text{product} &= a^2 - b^2 \\ &= (2\sqrt{11})^2 - (4\sqrt{15})^2 \\ &= 4 \times 11 - 16 \times 15 \\ &= 44 - 360 \\ &= -316\end{aligned}$$

f $\sqrt{a} + \sqrt{b}$

$$\begin{aligned}\text{product} &= x^2 - y^2 \\ &= \sqrt{a}^2 - \sqrt{b}^2 \\ &= a - b\end{aligned}$$

g $2\sqrt{m} - 3\sqrt{n}$

$$\begin{aligned}\text{product} &= a^2 - b^2 \\ &= (2\sqrt{m})^2 - (-3\sqrt{n})^2 \\ &= 4m - 9n\end{aligned}$$

h $a\sqrt{b} + c\sqrt{d}$

$$\begin{aligned}\text{product} &= x^2 - y^2 \\ &= (a\sqrt{b})^2 - (c\sqrt{d})^2 \\ &= a^2b - c^2d\end{aligned}$$

Page 19 questions

Rationalise the denominators of these fractions, and simplify as much as possible:

a $\frac{6}{\sqrt{2}}$

$$= \frac{6}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$= \frac{6\sqrt{2}}{2}$$

$$= 3\sqrt{2}$$

b $\frac{\sqrt{3}}{\sqrt{5}}$

$$= \frac{\sqrt{3}}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}}$$

$$= \frac{\sqrt{3} \times \sqrt{5}}{5}$$

$$= \frac{\sqrt{15}}{5}$$

c $\frac{3\sqrt{6}}{\sqrt{7}}$

$$= \frac{3\sqrt{6}}{\sqrt{7}} \times \frac{\sqrt{7}}{\sqrt{7}}$$

$$= \frac{3\sqrt{6} \times \sqrt{7}}{7}$$

$$= \frac{3\sqrt{42}}{7}$$

d $\frac{5}{2\sqrt{10}}$

$$= \frac{5}{2\sqrt{10}} \times \frac{\sqrt{10}}{\sqrt{10}}$$

$$= \frac{5\sqrt{10}}{2 \times 10}$$

$$= \frac{\sqrt{10}}{2 \times 2}$$

$$= \frac{\sqrt{10}}{4}$$

e $\frac{\sqrt{3} + 1}{\sqrt{12}}$

$$= \frac{\sqrt{3} + 1}{\sqrt{12}} \times \frac{\sqrt{12}}{\sqrt{12}}$$

$$= \frac{\sqrt{3} \times \sqrt{12} + \sqrt{12}}{12}$$

$$= \frac{\sqrt{36} + \sqrt{4 \times 3}}{12}$$

$$= \frac{6 + 2\sqrt{3}}{12}$$

$$= \frac{3 + \sqrt{3}}{6}$$

f $\frac{3\sqrt{12}}{\sqrt{6}}$

$$= \frac{3\sqrt{12}}{\sqrt{6}} \times \frac{\sqrt{6}}{\sqrt{6}}$$

$$= \frac{3\sqrt{72}}{6}$$

$$= \frac{\sqrt{36 \times 2}}{2}$$

$$= \frac{6\sqrt{2}}{2}$$

$$= 3\sqrt{2}$$

Page 19 questions

$$\begin{aligned}
 \text{g} \quad & \frac{4\sqrt{12} + \sqrt{2}}{\sqrt{2}} \\
 &= \frac{4\sqrt{12} + \sqrt{2}}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} \\
 &= \frac{4\sqrt{12} \times \sqrt{2} + \sqrt{2} \times \sqrt{2}}{2} \\
 &= \frac{4\sqrt{24} + 2}{2} \\
 &= 2\sqrt{4 \times 6} + 1 \\
 &= 4\sqrt{6} + 1
 \end{aligned}$$

$$\begin{aligned}
 \text{h} \quad & \frac{5\sqrt{7} + 3\sqrt{2}}{8\sqrt{5}} \\
 &= \frac{5\sqrt{7} + 3\sqrt{2}}{8\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} \\
 &= \frac{40\sqrt{35} + 24\sqrt{10}}{8 \times 5} \\
 &= \frac{5\sqrt{35} + 3\sqrt{10}}{5}
 \end{aligned}$$

$$\begin{aligned}
 \text{i} \quad & \frac{5}{\sqrt{m}} \\
 &= \frac{5}{\sqrt{m}} \times \frac{\sqrt{m}}{\sqrt{m}} \\
 &= \frac{5\sqrt{m}}{m}
 \end{aligned}$$

$$\begin{aligned}
 \text{j} \quad & \frac{2\sqrt{3x}}{7\sqrt{x}} \\
 &= \frac{2\sqrt{3x}}{7\sqrt{x}} \times \frac{\sqrt{x}}{\sqrt{x}} \\
 &= \frac{2\sqrt{3x} \times \sqrt{x}}{7x} \\
 &= \frac{2\sqrt{3}x}{7x} \\
 &= \frac{2\sqrt{3}}{7}
 \end{aligned}$$

Page 20 questions

Simplify these as much as possible, by rationalising the denominator of each fraction

$$\begin{aligned}
 \text{a} \quad & \frac{2}{\sqrt{3}} + \frac{3}{\sqrt{2}} \\
 &= \frac{2}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} + \frac{3}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} \\
 &= \frac{2\sqrt{3}}{3} + \frac{3\sqrt{2}}{2} \\
 &= \frac{2 \times 2\sqrt{3} + 3 \times 3\sqrt{2}}{6} \\
 &= \frac{4\sqrt{3} + 9\sqrt{2}}{6}
 \end{aligned}$$

$$\begin{aligned}
 \text{b} \quad & \frac{3}{\sqrt{7}} + \frac{\sqrt{6}}{\sqrt{5}} \\
 &= \frac{3}{\sqrt{7}} \times \frac{\sqrt{7}}{\sqrt{7}} + \frac{\sqrt{6}}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} \\
 &= \frac{3\sqrt{7}}{7} + \frac{\sqrt{30}}{5} \\
 &= \frac{5(3\sqrt{7}) + 7(\sqrt{30})}{35} \\
 &= \frac{15\sqrt{7} + 7\sqrt{30}}{35}
 \end{aligned}$$

Page 20 questions

$$\begin{aligned}
 \text{c } \frac{\sqrt{12}}{\sqrt{3}} - \frac{\sqrt{2}}{\sqrt{8}} &= \frac{\sqrt{12}}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} - \frac{1}{\sqrt{4}} \\
 &= \frac{\sqrt{36}}{3} - \frac{1}{2} \\
 &= \frac{6}{3} - \frac{1}{2} \\
 &= 2 - \frac{1}{2} \\
 &= \frac{3}{2}
 \end{aligned}$$

$$\begin{aligned}
 \text{d } \frac{5\sqrt{20}}{\sqrt{10}} + \frac{\sqrt{3}}{\sqrt{5}} &= \frac{5\sqrt{20}}{\sqrt{10}} \times \frac{\sqrt{10}}{\sqrt{10}} + \frac{\sqrt{3}}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} \\
 &= \frac{5\sqrt{200}}{10} + \frac{\sqrt{15}}{5} \\
 &= \frac{\sqrt{100 \times 2}}{2} + \frac{\sqrt{15}}{5} \\
 &= \frac{10\sqrt{2}}{2} + \frac{\sqrt{15}}{5} \\
 &= 5\sqrt{2} + \frac{\sqrt{15}}{5}
 \end{aligned}$$

Rationalise the denominator of the following:

$$\begin{aligned}
 \text{a } \frac{2}{\sqrt{2} + \sqrt{7}} &= \frac{2}{\sqrt{2} + \sqrt{7}} \times \frac{\sqrt{2} - \sqrt{7}}{\sqrt{2} - \sqrt{7}} \\
 &= \frac{2(\sqrt{2} - \sqrt{7})}{(\sqrt{2})^2 - (\sqrt{7})^2} \\
 &= \frac{2(\sqrt{2} - \sqrt{7})}{2 - 7} \\
 &= \frac{2(\sqrt{2} - \sqrt{7})}{-5} \\
 &= \frac{2(\sqrt{7} - \sqrt{2})}{5}
 \end{aligned}$$

$$\begin{aligned}
 \text{b } \frac{4\sqrt{6}}{2\sqrt{3} - \sqrt{5}} &= \frac{4\sqrt{6}}{2\sqrt{3} - \sqrt{5}} \times \frac{2\sqrt{3} + \sqrt{5}}{2\sqrt{3} + \sqrt{5}} \\
 &= \frac{8\sqrt{18} + 4\sqrt{30}}{(2\sqrt{3})^2 - (\sqrt{5})^2} \\
 &= \frac{8\sqrt{18} + 4\sqrt{30}}{4 \times 3 - 5} \\
 &= \frac{8\sqrt{18} + 4\sqrt{30}}{7}
 \end{aligned}$$

$$\begin{aligned}
 \text{c } \frac{2 - \sqrt{5}}{3\sqrt{7} + 2\sqrt{10}} &= \frac{2 - \sqrt{5}}{3\sqrt{7} + 2\sqrt{10}} \times \frac{3\sqrt{7} - 2\sqrt{10}}{3\sqrt{7} - 2\sqrt{10}} \\
 &= \frac{2(3\sqrt{7} - 2\sqrt{10}) - \sqrt{5}(3\sqrt{7} - 2\sqrt{10})}{(3\sqrt{7})^2 - (2\sqrt{10})^2} \\
 &= \frac{6\sqrt{7} - 4\sqrt{10} - 3\sqrt{35} + 2\sqrt{50}}{9 \times 7 - 4 \times 10} \\
 &= \frac{6\sqrt{7} - 4\sqrt{10} - 3\sqrt{35} + 2\sqrt{25 \times 2}}{63 - 40} \\
 &= \frac{6\sqrt{7} - 4\sqrt{10} - 3\sqrt{35} + 10\sqrt{2}}{23}
 \end{aligned}$$

$$\begin{aligned}
 \text{d } \frac{2\sqrt{3} + 3\sqrt{12}}{\sqrt{5} - 3\sqrt{2}} &= \frac{2\sqrt{3} + 3\sqrt{12}}{\sqrt{5} - 3\sqrt{2}} \times \frac{\sqrt{5} + 3\sqrt{2}}{\sqrt{5} + 3\sqrt{2}} \\
 &= \frac{2\sqrt{3}(\sqrt{5} + 3\sqrt{2}) + 3\sqrt{12}(\sqrt{5} + 3\sqrt{2})}{(\sqrt{5})^2 - (3\sqrt{2})^2} \\
 &= \frac{2\sqrt{15} + 6\sqrt{6} + 3\sqrt{60} + 9\sqrt{24}}{5 - 9 \times 2} \\
 &= \frac{2\sqrt{15} + 6\sqrt{6} + 3\sqrt{60} + 9\sqrt{4 \times 6}}{5 - 27} \\
 &= \frac{2\sqrt{15} + 6\sqrt{6} + 3\sqrt{60} + 18\sqrt{6}}{-22} \\
 &= -\frac{2\sqrt{15} + 24\sqrt{6} + 3\sqrt{60}}{22}
 \end{aligned}$$

Page 21 questions

Simplify the following as much as possible, by rationalising the denominator of each fraction:

$$\begin{aligned}
 \text{a } \frac{3}{\sqrt{5} + 2\sqrt{2}} + \frac{2}{\sqrt{7} + \sqrt{3}} &= \frac{3}{\sqrt{5} + 2\sqrt{2}} \times \frac{\sqrt{5} - 2\sqrt{2}}{\sqrt{5} - 2\sqrt{2}} + \frac{2}{\sqrt{7} + \sqrt{3}} \times \frac{\sqrt{7} - \sqrt{3}}{\sqrt{7} - \sqrt{3}} \\
 &= \frac{3\sqrt{5} + 6\sqrt{2}}{(\sqrt{5})^2 - (2\sqrt{2})^2} + \frac{2\sqrt{7} - 2\sqrt{3}}{(\sqrt{7})^2 - (\sqrt{3})^2} \\
 &= \frac{3\sqrt{5} + 6\sqrt{2}}{5 - 4 \times 2} + \frac{2\sqrt{7} - 2\sqrt{3}}{7 - 3} \\
 &= \frac{3\sqrt{5} + 6\sqrt{2}}{5 - 8} + \frac{2\sqrt{7} - 2\sqrt{3}}{4} \\
 &= \frac{3\sqrt{5} + 6\sqrt{2}}{-3} + \frac{\sqrt{7} - \sqrt{3}}{2} \\
 &= \frac{3\sqrt{5} + 6\sqrt{2}}{-3} \times \frac{-2}{-2} + \frac{\sqrt{7} - \sqrt{3}}{2} \times \frac{3}{3} \\
 &= \frac{-6\sqrt{5} - 12\sqrt{2}}{6} + \frac{3\sqrt{7} - 3\sqrt{3}}{6} \\
 &= \frac{-6\sqrt{5} - 12\sqrt{2} + 3\sqrt{7} - 3\sqrt{3}}{6} \\
 &= \frac{3\sqrt{7} - 3\sqrt{3} - 6\sqrt{5} - 12\sqrt{2}}{6} \\
 &= \frac{\sqrt{7} - \sqrt{3} - 2\sqrt{5} - 4\sqrt{2}}{2}
 \end{aligned}$$

$$\begin{aligned}
 \text{b } \frac{\sqrt{2}}{\sqrt{3} - \sqrt{10}} - \frac{1}{3\sqrt{5} - \sqrt{2}} &= \frac{\sqrt{2}}{\sqrt{3} - \sqrt{10}} \times \frac{\sqrt{3} + \sqrt{10}}{\sqrt{3} + \sqrt{10}} - \frac{1}{3\sqrt{5} - \sqrt{2}} \times \frac{3\sqrt{5} + \sqrt{2}}{3\sqrt{5} + \sqrt{2}} \\
 &= \frac{\sqrt{6} + \sqrt{20}}{(\sqrt{3})^2 - (\sqrt{10})^2} - \frac{3\sqrt{5} + \sqrt{2}}{(3\sqrt{5})^2 - (\sqrt{2})^2} \\
 &= \frac{\sqrt{6} + 2\sqrt{5}}{3 - 10} - \frac{3\sqrt{5} + \sqrt{2}}{45 - 2} \\
 &= \frac{\sqrt{6} + 2\sqrt{5}}{-7} - \frac{3\sqrt{5} + \sqrt{2}}{43} \\
 &= \frac{\sqrt{6} + 2\sqrt{5}}{-7} \times \frac{-43}{-43} - \frac{3\sqrt{5} + \sqrt{2}}{43} \times \frac{7}{7} \\
 &= \frac{43\sqrt{6} + 86\sqrt{5}}{301} - \frac{21\sqrt{5} + 7\sqrt{2}}{301} \\
 &= \frac{43\sqrt{6} + 86\sqrt{5} - 21\sqrt{5} - 7\sqrt{2}}{301} \\
 &= \frac{43\sqrt{6} + 65\sqrt{5} - 7\sqrt{2}}{301}
 \end{aligned}$$

Page 21 questions

Write the following in surd form

a $5^{\frac{1}{4}}$

$$= \sqrt[4]{5^1}$$

$$= \sqrt[4]{5}$$

b $3^{\frac{2}{3}}$

$$= \sqrt[3]{3^2}$$

$$= \sqrt[3]{9}$$

c $x^{\frac{4}{3}}$

$$= \sqrt[3]{x^4}$$

d $(5x)^{\frac{1}{2}}$

$$= \sqrt{5x}$$

e $100^{\frac{5}{7}}$

$$= \sqrt[7]{100^5}$$

$$= \sqrt[7]{10,000,000,000}$$

f $(4q)^{\frac{7}{4}}$

$$= \sqrt[4]{(4q)^7}$$

Write the following index form:

a $\sqrt{7}$

$$= 7^{\frac{1}{2}}$$

b $\sqrt{7^3}$

$$= 7^{\frac{3}{2}}$$

c $\sqrt[3]{5}$

$$= 5^{\frac{1}{3}}$$

d $\sqrt[6]{3^8}$

$$= 3^{\frac{8}{6}}$$

$$= 3^{\frac{4}{3}}$$

e $\sqrt{y^5}$

$$= y^{\frac{5}{2}}$$

f $\sqrt[3]{p^4}$

$$= p^{\frac{4}{3}}$$

g $\sqrt[n]{4}$

$$= 4^{\frac{1}{n}}$$

h $\sqrt[p]{x^q}$

$$= x^{\frac{q}{p}}$$

Page 22 questions

Find the value of the following (without a calculator)

a $4^{\frac{1}{2}}$

$$= \sqrt{4}$$

$$= 2$$

c $16^{\frac{3}{2}}$

$$\text{either } = (\sqrt{16})^3$$

$$= 4^3$$

$$= 64$$

$$\text{Or } = \sqrt{16^3}$$

$$= \sqrt{4096}$$

$$= 64$$

e $125^{\frac{2}{3}}$

$$= (\sqrt[3]{125})^2$$

$$= 5^2$$

$$= 25$$

g $10000^{\frac{3}{4}}$

$$= \sqrt[4]{10000^3}$$

$$= (\sqrt[4]{10000})^3$$

$$= (10)^3$$

$$= 1000$$

i $\left(\frac{8}{343}\right)^{\frac{1}{3}}$

$$= \sqrt[3]{\frac{8}{343}}$$

$$= \frac{\sqrt[3]{8}}{\sqrt[3]{343}}$$

$$= \frac{\sqrt[3]{2 \times 2 \times 2}}{\sqrt[3]{7 \times 7 \times 7}}$$

$$= \frac{2}{7}$$

b $27^{\frac{1}{3}}$

$$= \sqrt[3]{27}$$

$$= \sqrt[3]{3 \times 3 \times 3}$$

$$= 3$$

d $9^{\frac{3}{2}}$

$$= (\sqrt{9})^3$$

$$= 3^3$$

$$= 3 \times 3 \times 3$$

$$= 27$$

f $100^{\frac{5}{2}}$

$$= (\sqrt{100})^5$$

$$= 10^5$$

$$= 10 \times 10 \times 10 \times 10 \times 10$$

$$= 10,000$$

h $64^{\frac{1}{6}} \times 32^{\frac{1}{5}}$

$$= \sqrt[6]{64} \times \sqrt[5]{32}$$

$$= \sqrt[6]{2 \times 2 \times 2 \times 2 \times 2 \times 2} \times \sqrt[5]{2 \times 2 \times 2 \times 2 \times 2}$$

$$= 2 \times 2$$

$$= 4$$

j $(27^{\frac{2}{3}} \times 32^{\frac{2}{5}})^{\frac{1}{2}}$

$$= \sqrt{27^{\frac{2}{3}} \times 32^{\frac{2}{5}}}$$

$$= \sqrt{(\sqrt[3]{27})^2 \times (\sqrt[5]{32})^2}$$

$$= \sqrt{(\sqrt[3]{27} \times \sqrt[5]{32})^2}$$

$$= \sqrt[3]{27} \times \sqrt[5]{32}$$

$$= 3 \times 2$$

$$= 6$$

Page 22 questions

Show that $\frac{2}{\sqrt{7}+4} - \frac{2}{\sqrt{7}-4}$ is a rational number.

A rational number is one where that can be written numerator and denominator that are both integers, and where the denominator is not zero.

If we can simplify the expression and show the numerator and denominator are both integers, then we show it is a rational number.

Start by rationalising the denominator of each fraction and see what happens.

$$\begin{aligned}\frac{2}{\sqrt{7}+4} - \frac{2}{\sqrt{7}-4} &= \frac{2}{\sqrt{7}+4} \times \frac{\sqrt{7}-4}{\sqrt{7}-4} - \frac{2}{\sqrt{7}-4} \times \frac{\sqrt{7}+4}{\sqrt{7}+4} \\ &= \frac{2\sqrt{7}-8}{7-16} - \frac{2\sqrt{7}+8}{7-16} \\ &= \frac{-16}{7-16} \\ &= \frac{-16}{-9} \\ &= \frac{16}{9}\end{aligned}$$

∴ This is a rational number

